Goldman Sachs Research

Equity Research

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TAIWAN CCL

July revenue trending much better than our bullish estimate on high-end CCL players, with solid pricing uptrend; Buy EMC/TUC with ~100% upside

Chao Wang
+886(2)2730-4195 | kuan-chao.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

Allen Chang
+852-2978-2930 | allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

Al Wang
+886(2)2730-4081 | al.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

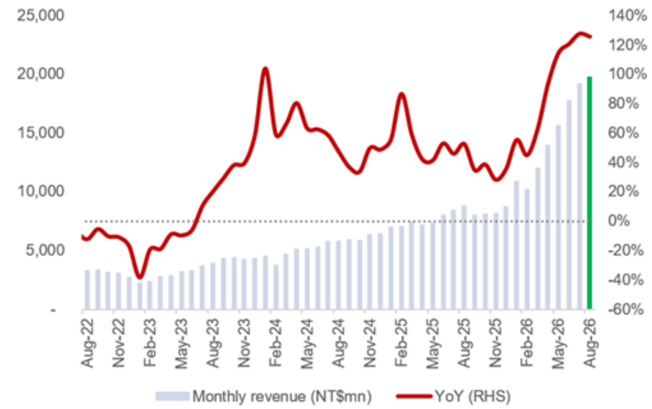
July revenue review - EMC/TUC revenue grew +8%/+21% MoM, which was 3/5% higher than our Street most bullish estimate

EMC announced its July revenue at NT$19,207mn (+8% MoM, +128% YoY), which was **3% higher than our estimate (most bullish on the street; our 3Q revenue estimate is 17% higher than BBG consensus already)**, and accounted for 35% of the company's conservative 3Q revenue guidance (see here), thanks to the much better than expected high-end AI product demand (from Trainium 3 and Rubin AI servers) and the improving pricing level (see more here). Considering the demand outlook is likely to continue to be stronger every month in the rest of the year (with more high-end AI projects ramping up), we believe the company's August and September revenue will further go up with better product mix and pricing, implying the 3Q revenue to go up by at least 22% QoQ (vs. company's guidance/ GSe at +15%/+21 QoQ). **For August, we expect EMC's revenue to further go up to NT$19.8bn (+3% QoQ, +126% YoY)**, with the solid improving product mix.

TUC also announced its July revenue at NT$5,908mn (+22% MoM, +98% YoY), which was **5% higher than our estimate (most bullish on the street; our 3Q revenue estimate is ~10% higher than BBG consensus)**, and we believe the solid beat on the July revenue was driven by (1) the improving pricing trend every month (we also see the M7 grade CCL pricing starting to go up from middle of July), and (2) the improving contribution from new Thailand capacity with solid demand from ASIC customers, while TUC's solid outgrowth vs. EMC in July revenue was due to (1) its Thailand plant ramping suggesting a better market share in the ASIC supply chain, and (2) TUC's more proactive pricing strategy for M7 & below grade CCL vs. EMC. **For August, we expect TUC's revenue to further go up to NT$6.2bn (+5% QoQ, +144% YoY)**, considering an even better pricing outlook (more pricing hike on high-end and low-end CCL) and the increasing contribution from new Thailand capacity (~10% of total capacity started to ramp up from middle of July, and the contribution will double MoM in August with 2x more working days).

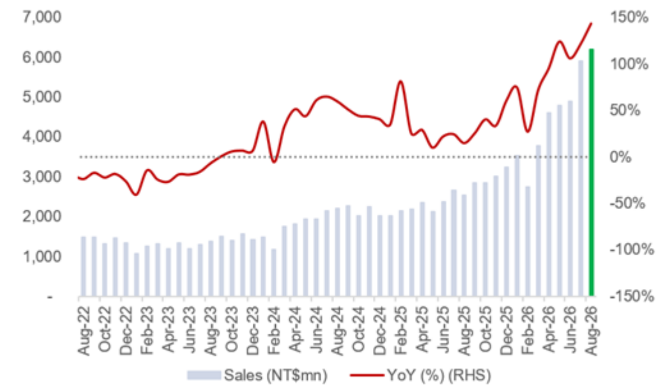
Maintain Buy on EMC and TUC with unchanged TP of NT$10,200 / NT$3,010, suggesting ~100% upside vs. today's close.

Exhibit 1: We estimate EMC's August revenue to increase by 126% YoY



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: We estimate TUC's August revenue to increase by 144% YoY



Source: Company data, Goldman Sachs Global Investment Research

Investment view

We continue to see a solid S/D outlook on the high-end / high quality CCL products in coming years (see more details here), and expect the CCL technology migration will start to accelerate from 2H26 (while EMC just started the shipment of the most high end M9 grade CCL in the market) which should suggest an even better order demand/ product mix, as well as pricing level in coming quarters. The solid beat on our already bullish July revenue suggesting an even stronger pricing condition and customers' high-end AI CCL demand going into the peak quarter (Trainium 3 order will peak in 3Q, while Rubin and TPU 8 demand will start ramping up from 3Q26 to 4Q26). We believe EMC as the key AI ASIC player will further enjoy even better demand/ product mix, while TUC, who focus on Trianium and high-end 800G/1.6T swtich, will also see a good product mix /pricing outlook in the rest of the year.

Overall speaking, we still like the high-end CCL supply chain the most in the TW PCB supply chain, especially those CCL companies who already proved to have good product quality/ production capability on M7/8/9 grade CCL (e.g. EMC / TUC), and expect the next round of high-end CCL margin expansion just started from 2Q26, which will at least continue in 2H26-1H27 (see here). We believe we are still in the early stage of the high-end CCL pricing hike cycle and only seeing the early demand growth of the overall high-end CCL TAM outlook in coming years (we expect AI CCL TAM will grow by 250%+ 2025-28E CAGR - here), and expect the key high-end players' EMC and TUC will be benefited from the trend the most.

Valuation and risk

EMC

We maintain Buy on EMC, with an unchanged TP of NT$10,200 is based on 27x 2H27-1H28E P/E (in line with the peak P/E multiple for AI component suppliers in the past 3 years).

Exhibit 3: EMC P&L table

NT$mn	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E	2029E
Revenue	33,067	47,275	57,349	59,952	67,047	81,696	94,459	109,822	122,256	139,528	153,104	165,161	94,261	197,643	353,024	580,049	827,957
Gross profit	9,729	16,003	21,117	22,649	26,204	32,690	38,502	46,151	52,156	60,074	66,695	72,969	28,120	69,498	143,548	251,894	374,578
Operating expense	(2,601)	(3,269)	(3,556)	(3,897)	(4,392)	(4,616)	(4,912)	(5,162)	(5,502)	(6,000)	(6,277)	(6,606)	(9,012)	(13,322)	(19,081)	(24,385)	(34,812)
Operating income	7,128	12,734	17,561	18,752	21,813	28,075	33,591	40,990	46,655	54,074	60,418	66,362	19,108	56,176	124,467	227,509	339,766
Pretax income	7,194	12,785	17,561	18,752	21,813	28,075	33,591	40,990	46,655	54,074	60,418	66,362	18,876	56,293	124,467	227,509	339,766
Taxes expense	(1,855)	(2,914)	(4,039)	(4,126)	(5,017)	(6,457)	(7,726)	(9,428)	(10,731)	(12,437)	(13,896)	(15,263)	(4,231)	(12,934)	(28,628)	(52,327)	(78,146)
Net income	5,340	9,873	13,522	14,627	16,796	21,618	25,865	31,562	35,924	41,637	46,522	51,099	14,649	43,362	95,840	175,182	261,620
EPS, NT$	**14.90**	**27.55**	**37.73**	**40.81**	**46.87**	**60.32**	**72.17**	**88.07**	**100.24**	**116.18**	**129.81**	**142.58**	**41.67**	**121.00**	**267.42**	**488.81**	**730.00**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	29.4%	33.9%	36.8%	37.8%	39.1%	40.0%	40.8%	42.0%	42.7%	43.1%	43.6%	44.2%	29.8%	35.2%	40.7%	43.4%	45.2%
Operating expense ratio	7.9%	6.9%	6.2%	6.5%	6.6%	5.7%	5.2%	4.7%	4.5%	4.3%	4.1%	4.0%	9.6%	6.7%	5.4%	4.2%	4.2%
Operating margin	21.6%	26.9%	30.6%	31.3%	32.5%	34.4%	35.6%	37.3%	38.2%	38.8%	39.5%	40.2%	20.3%	28.4%	35.3%	39.2%	41.0%
Net margin	16.1%	20.9%	23.6%	24.4%	25.1%	26.5%	27.4%	28.7%	29.4%	29.8%	30.4%	30.9%	15.5%	21.9%	27.1%	30.2%	31.6%
QoQ growth (%)																	
Revenue	32.7%	43.0%	21.3%	4.5%	11.8%	21.8%	15.6%	16.3%	11.3%	14.1%	9.7%	7.9%	-	-	-	-	-
Gross profit	36.6%	64.5%	32.0%	7.3%	15.7%	24.8%	17.8%	19.9%	13.0%	15.2%	11.0%	9.4%	-	-	-	-	-
Operating income	44.5%	78.6%	37.9%	6.8%	16.3%	28.7%	19.6%	22.0%	13.8%	15.9%	11.7%	9.8%	-	-	-	-	-
Net income	42.9%	84.9%	37.0%	8.2%	14.8%	28.7%	19.6%	22.0%	13.8%	15.9%	11.7%	9.8%	-	-	-	-	-
YoY growth (%)																	
Revenue	52.5%	110.0%	128.1%	140.5%	102.8%	72.8%	64.7%	83.2%	82.3%	70.8%	62.1%	50.4%	46%	110%	79%	64%	43%
Gross profit	47.6%	134.3%	178.7%	217.9%	169.3%	104.3%	82.3%	103.8%	99.0%	83.8%	73.2%	58.1%	56%	147%	107%	75%	49%
Operating income	57.0%	174.2%	251.9%	280.0%	206.0%	120.5%	91.3%	118.6%	113.9%	92.6%	79.9%	61.9%	57%	194%	122%	83%	49%
Net income	53.9%	183.9%	241.0%	291.4%	214.5%	118.9%	91.3%	115.8%	113.9%	92.6%	79.9%	61.9%	53%	196%	121%	83%	49%

Source: Company data, Goldman Sachs Global Investment Research

TUC

We maintain Buy on TUC, with an unchanged TP of NT$3,010 is based on a 22x 2027E P/E (+2x STDV higher than the past 3-year industry average P/E multiple).

Exhibit 4: TUC P&L table

NT$mn	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E	2029E
Revenue	10,054	14,301	17,621	19,785	21,664	25,821	33,587	39,348	41,153	45,886	55,305	60,569	30,340	61,760	120,421	202,913	297,071
Gross profit	2,529	4,250	5,835	6,897	7,712	9,395	12,468	14,883	15,466	17,196	21,093	23,533	6,898	19,510	44,457	77,288	117,670
Operating expense	(701)	(892)	(1,022)	(1,108)	(1,181)	(1,381)	(1,511)	(1,613)	(1,708)	(1,790)	(1,963)	(2,271)	(2,554)	(3,723)	(5,687)	(7,732)	(11,341)
Operating income	1,828	3,358	4,813	5,789	6,531	8,013	10,956	13,270	13,758	15,407	19,130	21,262	4,344	15,787	38,770	69,556	106,329
Pretax income	1,874	3,427	4,813	5,269	6,011	7,613	10,956	13,270	13,722	15,372	19,130	21,262	4,519	15,383	37,850	69,485	106,258
Taxes expense	(614)	(1,085)	(1,155)	(1,265)	(1,503)	(1,903)	(2,739)	(3,317)	(3,431)	(3,843)	(4,782)	(5,315)	(1,109)	(4,119)	(9,463)	(17,371)	(26,565)
Net income	1,260	2,342	3,658	4,005	4,508	5,710	8,217	9,952	10,292	11,529	14,347	15,946	3,409	11,264	28,388	52,114	79,694
EPS, NT$	**4.36**	**8.02**	**12.53**	**13.72**	**15.44**	**19.56**	**28.14**	**34.08**	**35.25**	**39.48**	**49.14**	**54.61**	**12.13**	**38.63**	**97.22**	**178.48**	**272.94**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	25.1%	29.7%	33.1%	34.9%	35.6%	36.4%	37.1%	37.8%	37.6%	37.5%	38.1%	38.9%	22.7%	31.6%	36.9%	38.1%	39.6%
Operating expense ratio	7.0%	6.2%	5.8%	5.6%	5.5%	5.4%	4.5%	4.1%	4.2%	3.9%	3.6%	3.8%	8.4%	6.0%	4.7%	3.8%	3.8%
Operating margin	18.2%	23.5%	27.3%	29.3%	30.1%	31.0%	32.6%	33.7%	33.4%	33.6%	34.6%	35.1%	14.3%	25.6%	32.2%	34.3%	35.8%
Net margin	12.5%	16.4%	20.8%	20.2%	20.8%	22.1%	24.5%	25.3%	25.0%	25.1%	25.9%	26.3%	11.2%	18.2%	23.6%	25.7%	26.8%
QoQ growth (%)																	
Revenue	10.2%	42.2%	23.2%	12.3%	9.5%	19.2%	30.1%	17.2%	4.6%	11.5%	20.5%	9.5%	-	-	-	-	-
Gross profit	25.5%	68.1%	37.3%	18.2%	11.8%	21.8%	32.7%	19.4%	3.9%	11.2%	22.7%	11.6%	-	-	-	-	-
Operating income	41.7%	83.7%	43.3%	20.3%	12.8%	22.7%	36.7%	21.1%	3.7%	12.0%	24.2%	11.1%	-	-	-	-	-
Net income	16.4%	85.8%	56.2%	9.5%	12.6%	26.6%	43.9%	21.1%	3.4%	12.0%	24.4%	11.1%	-	-	-	-	-
YoY growth (%)																	
Revenue	57.8%	110.9%	118.5%	116.8%	115.5%	80.6%	90.6%	98.9%	90.0%	77.7%	64.7%	53.9%	32%	104%	95%	69%	46%
Gross profit	64.9%	196.1%	204.7%	242.3%	205.0%	121.1%	113.7%	115.8%	100.5%	83.0%	69.2%	58.1%	29%	183%	128%	74%	52%
Operating income	95.3%	293.9%	280.3%	348.9%	257.4%	138.6%	127.7%	129.2%	110.7%	92.3%	74.6%	60.2%	30%	263%	146%	79%	53%
Net income	87.4%	259.3%	264.8%	270.0%	257.8%	143.8%	124.7%	148.5%	128.3%	101.9%	74.6%	60.2%	31%	230%	152%	84%	53%

Source: Company data, Goldman Sachs Global Investment Research

Investment Thesis & PT methodology and risks

EMC, a key supplier of high-end HDI material (70%+ market share) and SLP material (90%+ market share), has been focusing on the high-speed switch/server CCL market for 3-5+ years. We are positive on its market share trajectory in the server industry and expect EMC to increase its share from <10% in the Purley generation to 15-20%/20+% in the Whitley/Eagle Stream platforms. EMC is also proactively expanding its share in the switch market (40%/30%+ in 2024/23 vs. <5% before 2019), with an expanding AI customer base (only one 400G customer in 3Q20, to being one of the largest suppliers in 2023), which should continue to benefit the company's top/bottom line growth over time. We believe EMC's leading position in the AI server CCL market should drive strong revenue growth and GM/OPM expansion in the long term, given that it is the major supplier for all Nvidia/Google/AWS AI server projects, and we expect it to maintain its leadership position. Considering its leading position in the high-end CCL industry, solid production skills and earnings growth outlook, we are Buy rated on the stock. With the shares trading at a 2026E P/E below AI server component suppliers and supported by a solid growth outlook, we view valuation as attractive.

Key downside risks include: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and (3) rising competition from mainland China peers.

Valuation methodology: Our 12m TP of NT$10,200 is based on 27x 2H27-1H28E P/E (in line with the peak P/E multiple for AI component suppliers in the past 3 years).

Key downside risks: (1) RCC to replace all high-end smartphone HDI design; (2) rising trade tensions, which could lead to weaker smartphone and server shipments; and (3) rising competition from mainland China peers.

TUC is a key high-end global CCL supplier that focuses on M7+ grade high-speed CCL (key applications include high-end switches (100G+) and AI server materials), with 20%+ market share in the past two years. The company continues to expect its unit market share in the server industry to increase from 15% in Whitley and Purley processors to higher in the Eagle Stream generation, and plans to launch two types of p roducts for low-end and high-end customers (T2A and T2C) to gain market share. Also, strong growth in 400G/800G switch shipments and new 800G switch shipments, scheduled for launch in 2025-27, should continue to be a key demand driver for TUC, in our view. Additionally, the company is working on high-end AI projects with CSP and enterprise players, which we believe should drive revenue momentum in the long term. We view the stock as undervalued vs. Taiwan CCL peers on a P/E basis and rate it Buy.

Key downside risks include: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

Valuation methodology: Our 12m TP of NT$3,010 is based on a 22x 2027E P/E (+2x STDV higher than the past 3-year industry average P/E multiple).

Key downside risks: (1) slower-than-expected share gains in the low-loss CCL segment; (2) rising trade tensions, which could lead to weaker server and switch shipments globally; and (3) rising competition from mainland China peers.

Disclosure Appendix

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I, Chao Wang, hereby certify that all of the views expressed in this report accurately reflect my personal views about the subject company or companies and its or their securities. I also certify that no part of my compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chao Wang Goldman Sachs (Asia) L.L.C., Taipei Branch, Allen Chang Goldman Sachs (Asia) L.L.C., Al Wang Goldman Sachs (Asia) L.L.C., Taipei Branch.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

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Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Elite Material and Taiwan Union Technology Corp. is/are relative to the other companies in its/their coverage universe: AWSC, Airtac International Group, Co-Tech Development Corp., Delta Electronics, Elite Material, GCE, Hiwin Corp., ITEQ Corp, Kinsus, Lotes, NYPCB, Taiwan Union Technology Corp., Unimicron Technology, Win Semiconductors Corp., Yageo Corp., Zhen Ding Technology Holding

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the second most recent month end: Elite Material (NT$5,245.00) and Taiwan Union Technology Corp. (NT$1,470.00)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Elite Material (NT$5,245.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Elite Material (NT$5,245.00)

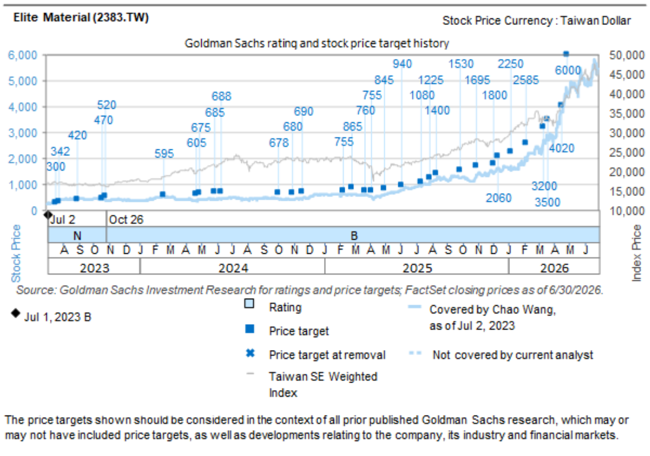
Distribution of ratings/investment banking relationships

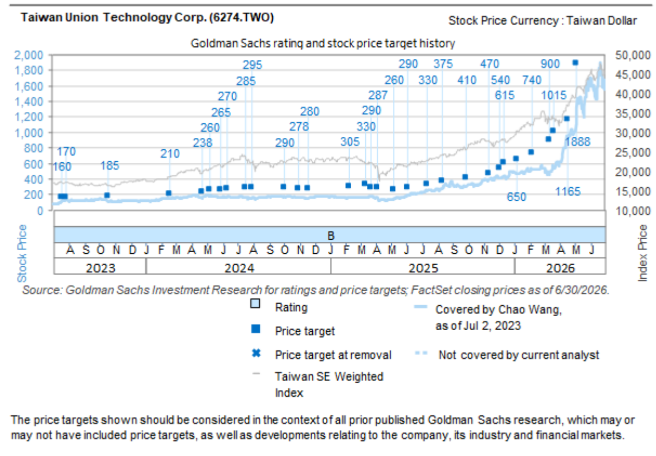
Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)





Target price history table(s)

Taiwan Union Technology Corp. (6274.TWO)

Date of report	Target price (NT$)	Closing price (NT$)
29-Jul-26	3,010.00	1,020.00
17-Jul-26	2,860.00	1,170.00
06-May-26	1,888.00	1,315.00
17-Apr-26	1,165.00	946.00
20-Mar-26	1,015.00	554.00
12-Mar-26	900.00	452.50
05-Feb-26	740.00	520.00
06-Jan-26	650.00	476.00
12-Dec-25	615.00	443.00
04-Dec-25	540.00	411.00
12-Nov-25	470.00	407.00
29-Sep-25	410.00	298.00
13-Aug-25	375.00	282.50
15-Jul-25	330.00	264.00
05-Jun-25	290.00	188.50
07-May-25	260.00	142.50
06-Apr-25	287.00	159.00
24-Mar-25	290.00	176.50
12-Mar-25	330.00	160.50
09-Feb-25	305.00	163.50
20-Nov-24	280.00	155.00
30-Oct-24	278.00	162.00
03-Oct-24	290.00	175.00
31-Jul-24	295.00	160.00
17-Jul-24	285.00	168.00
11-Jun-24	270.00	164.50
29-May-24	265.00	167.00
08-May-24	260.00	185.50
23-Apr-24	238.00	164.50
18-Feb-24	210.00	156.50
20-Oct-23	185.00	111.00

Elite Material (2383.TW)

Date of report	Target price (NT$)	Closing price (NT$)
29-Jul-26	10,200.00	4,100.00
17-Jul-26	9,500.00	4,495.00
29-Apr-26	6,000.00	4,440.00
17-Apr-26	4,020.00	3,800.00
20-Mar-26	3,500.00	2,810.00
12-Mar-26	3,200.00	2,600.00
05-Feb-26	2,585.00	1,950.00
06-Jan-26	2,250.00	1,695.00
12-Dec-25	2,060.00	1,605.00
04-Dec-25	1,800.00	1,425.00
30-Oct-25	1,695.00	1,275.00
29-Sep-25	1,530.00	1,225.00
13-Aug-25	1,400.00	1,200.00
30-Jul-25	1,225.00	1,005.00
15-Jul-25	1,080.00	970.00
05-Jun-25	940.00	789.00
30-Apr-25	845.00	556.00
06-Apr-25	755.00	527.00
24-Mar-25	760.00	604.00
25-Feb-25	865.00	576.00
09-Feb-25	755.00	615.00
20-Nov-24	690.00	447.50
30-Oct-24	680.00	419.50
03-Oct-24	678.00	442.00
11-Jun-24	688.00	422.50
29-May-24	685.00	427.00
30-Apr-24	675.00	412.00
23-Apr-24	605.00	376.00
18-Feb-24	595.00	534.00
26-Oct-23	520.00	388.50
20-Oct-23	470.00	403.00
30-Aug-23	420.00	415.50

Price targets shown in table(s) are unadjusted for corporate actions.

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